HIERARCHICAL LEARNING FOR THE UNSUPERVISED SEGMENTATION OF RADAR SOUNDER DATA ACQUIRED ON THE CRYOSPHERE

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ABSTRACT

In the radar sounder literature, extracting subsurface geological information relies on supervised deep learning with large labeled datasets. While some methods reduce the need for extensive labels through weak supervision, there remains a gap in the availability of unsupervised segmentation techniques. This paper proposes a novel method for unsupervised radargram segmentation based on incremental learning (IL). The method involves the prior geophysical modeling of the cryosphere subsurface targets into a class hierarchy. Through several IL steps, a network is trained to progressively extract semantically meaningful features that are analyzed to compute the segmentation map. Each step refines the segmentation map by considering the new targets from the following level of the class hierarchy that details the targets at the previous level. To enhance the training process, contrastive learning is incorporated, along with techniques for distilling information from prior iterations to recall the network the properties of previously seen classes. To validate the effectiveness of the proposed method, we conducted successful experiments on MCoRDS-3 data acquired in Greenland.

***Index Terms*—** Unsupervised semantic segmentation, contrastive learning, incremental learning, image processing, radar sounder.

1. INTRODUCTION

Radar sounder data, called radargrams, unveil unique information on subsurface geology. Recent methods (e.g., [1, 2]) utilize machine and deep learning techniques to segment radargrams, automatically extracting information on subsurface geological targets. This approach overcomes the limitations of manual radargram analysis. Less recent works automatically analyzed features manually designed to consider the geometrical and radiometric properties of the targets [3, 4]. Classifiers, such as Support Vector Machines (SVM), have been used to label each pixel of the radargram independently. More recent works utilize deep convolutional neural networks (CNNs) to extract semantically meaningful features from radargrams automatically [1, 5]. CNNs exploit convolutional filters that analyze the pixel and its surroundings to automatically embed information about the spatial relation between the subsurface targets. These works are supervised, and they assume the availability of a large labeled dataset to avoid network overfitting and ensure reliable training. Literature works adopt mitigation techniques to achieve reliable training with weak supervision and reduce the number of labeled samples [1, 5]. However, labeled datasets in the radar sounder domain require detailed subsurface knowledge that is hardly available. Another limitation is that the training is based on losses (such as cross-entropy loss) computed pixel-wise, which ignores the spatial interdependence between pixels.

Recently, attempts have been made to design methods with unsupervised network training to segment radargrams [6]. The method [6] automatically extracts and selects semantically meaningful features. Then, for each class, the feature vector of a labeled anchor pixel is used as a reference to cluster the pixels belonging to the same class. Although the method requires a very small amount of labeled data, there is a need for a novel method to segment radargrams in a completely unsupervised manner and without requiring information on the subsurface geology.

This paper extends the work in [6] and proposes a novel method for the unsupervised segmentation of radargrams acquired in the cryosphere, i.e., assigning a label to each pixel to identify subsurface targets without any labeled data. The approach employs incremental learning (IL) and is based on prior geophysical modeling of cryosphere subsurface targets within a class hierarchy. Over multiple IL steps, the network is trained to extract semantically meaningful features, which are then analyzed to generate the segmentation map. Each iteration introduces new classes that refine the target segmentation at the previous level, taking into account the targets/classes at the next level of the geophysical hierarchy. To enhance the training process, we incorporate contrastive learning and distillation. The first models the semantic relationship between pixels to learn an embedding space by pulling the representations of the same target pixels close and pushing those of different targets apart. We also distill the information from prior iterations to recall the network the properties of previously seen classes. The proposed method was successfully validated by segmenting radargrams acquired in Greenland by MCoRDS-3.

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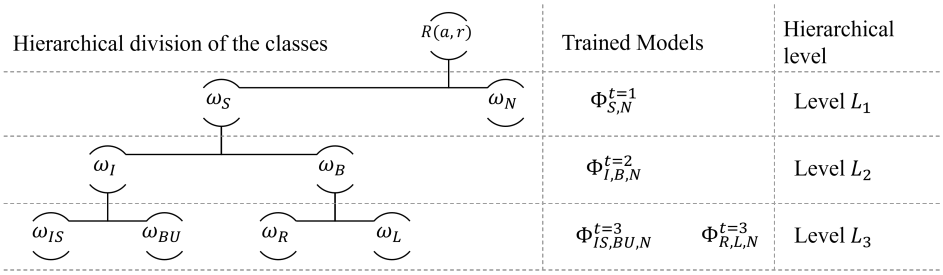


Fig. 1. Class hierarchy of the subsurface targets in cryosphere radargrams.

2. PROPOSED METHOD

A radargram $R(a, r)$ contains reflections from the surface and subsurface (signal pixels belonging to class ω_S) surrounded by the noise-limited regions (no-signal pixels belonging to class ω_N). Leveraging on the prior knowledge of the target distribution in the terrestrial cryosphere subsurface [1], we can assume that the target pixels can be clustered into two classes (see Fig. 1)– icy targets and basal interface belonging to classes ω_I and ω_B, respectively. ω_I and ω_B can be hierarchically subdivided into more detailed targets. Icy targets (ω_I) can be divided into the ice stratigraphy (ω_{IS}) and the basal ice (ω_{BU}) at the bottom of the ice sheet, and the basal interface (ω_B) can be rocky (ω_R) or show water parts (ω_L).

We propose an iterative learning method that leverages the illustrated hierarchy of subsurface targets (Fig. 1). The method iteratively segments radargrams, starting from the signal and noise-limited regions and adding new classes at each incremental learning step $t, t = [1, N_t]$, where $N_t = \Omega - 1$, and Ω is the total number of classes. The core idea is that each step t solves the classification problem defined at the corresponding level of class hierarchy (see Fig. 1). Hence, we divide the radargram segmentation into easier tasks $\mathcal{T}_t$ where the model $\Phi^t(\cdot)$ discriminates between the two classes. At $t = 1$, we aim at discriminating between ω_S and ω_N (level 1 of the class hierarchy). At $t > 1$, we aim to discriminate between pixels belonging to the class at a deeper hierarchical level. Here, we update the label of pixels in ω_S to identify different subsurface targets (e.g., ice sheet vs. basal interface at $t = 2$).

At each incremental learning step t, we train a deep network (called encoder) to extract a set of N_F features $\Phi^t(R^t(a, r))$ (see Fig. 2) considering a proxy task (different than radargram segmentation) and contrastive training loss [7]. $\Phi^t(R^t(a, r))$ are analyzed to select the most meaningful features $F^t(a, r, f)$ considering the structural similarity index measure (SSMI) [6]. The segmentation map $\mathcal{M}(a, r)^t$ is computed with the argmax operation over the selected features:

$$\mathcal{M}^t(a, r) = \text{argmax}(F^t(a, r, f)). \tag{1}$$

2.1. Training strategy

We apply a common self-supervised strategy by feeding the network with a radargram and its augmented version [8]. Training with a contrastive loss, the network learns that visually distinct inputs share the same information and image the same targets. The contrastive loss models the semantic relationship between adjacent pixels and learns an embedding space by pushing close representations for pixels of the same class (subsurface target) and far representations for those of different classes (targets). Compared with commonly used losses (such as the cross-entropy [1, 5]), the contrastive loss forces the network to learn the semantic relationship between adjacent pixels.

As contrastive logic, we consider the momentum contrast [7] that transforms the training problem into a dictionary look-up problem. Each input radargram $\mathbf{R}_q$ is associated with a query q, generated by encoding $\mathbf{R}_q$ with the encoder. The query is compared with a large consistent dictionary of keys $k_i, i \in [0, K]$ (see Fig. 3). To calculate the distance between the query q and a specific key k_i, we consider the dot product and the contrastive function InfoNCE [7]:

$$\mathcal{L} = -\log \frac{\exp(q \cdot k_i/\tau)}{\sum_{i=0}^{K} \exp(q \cdot k_i/\tau)} \tag{2}$$

where τ is a temperature hyper-parameter. InfoNCE estimates the mutual information between the q and k_i that forces features of a radargram and those of the augmented version to be close and those of unrelated radargrams to be distant. While an explicit dictionary is absent for RS data, it is dynamically computed at each training step using a momentum encoder. Encoded data are updated, maintaining consistency with new keys and removing outdated entries. Given the large dimension of the dictionary, it is impossible to update the momentum encoder with back-propagation as the gradient should propagate through the queue. [7] proposes to update the momentum encoder using a momentum coefficient $m \in [0, 1]$ defined as:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q \tag{3}$$

where θ_k represents the parameters of the momentum encoder, and θ_q those of the other encoder. θ_q are updated

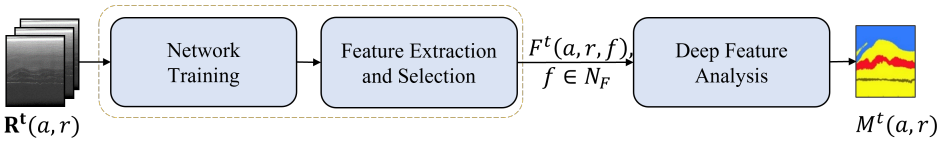


Fig. 2. Block diagram for the proposed method for each step k.

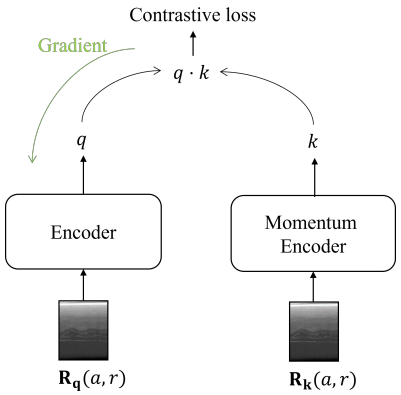


Fig. 3. Schematic representation of the Momentum Contrast logic (Fig. adapted from [7]).

considering back-propagation, while θ_k are updated more smoothly according to coefficient m.

To mitigate the propagation of segmentation errors from the previous step, in the inference phase, the network discriminates between the classes at the current level t and those at the previous step $t-1$. For example, model $\Phi^t = 2_{I,B,N}$ segments the radargrams into the two signal classes ω_I and ω_B and the noise-limited regions class ω_N. Although incremental learning shows promising performances in several computer vision applications, it is affected by the well-known problem of catastrophic forgetting. The model cannot continuously be updated with new data without losing the previously learned knowledge [9]. As done in the literature [10], we rely on distilling the knowledge learned in the previous step $t-1$ in the current step t by enforcing the predictions of the network on the old task not to fluctuate much while updating the parameters with the new task. To this end, we mask the training radargrams at t by hiding the pixels labeled as background in the previous step (e.g., ω_N in $t = 2$).

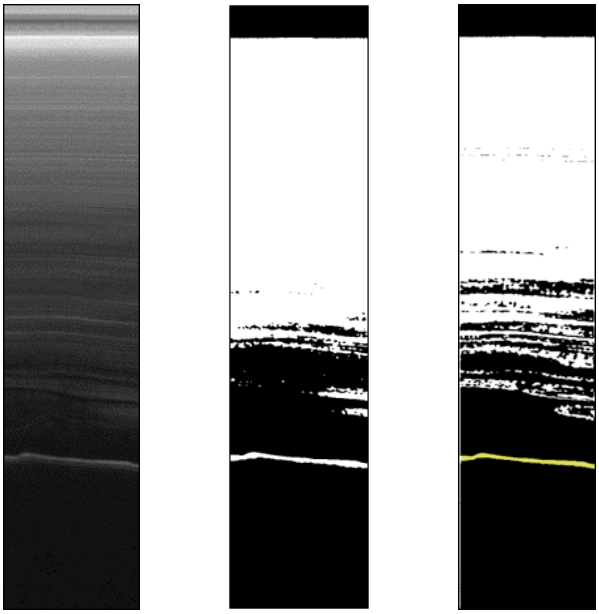
2.2. Network architecture

The architecture of the two encoders is the same. Similarly to [6], each encoder consists of an autoencoder backbone followed by a contracting path. The backbone has an autoencoder shape to extract N_F features with the same spatial dimension as the input radargrams. The contracting path extracts the query and the keys for the momentum contrast training. As a contracting path, we propose five convolutional blocks to reduce the dimension of the features smoothly. Each block consists of a 3x3 convolutional layer, a normalization layer, a ReLu activation function, a 1x1 convolutional layer, a normalization layer, and a ReLu activation. A 2x2 max-pooling layer follows each block to reduce the dimension of the extracted features. We apply an adaptive average pooling layer to reduce further the feature dimension that adaptively averages the input features and a flatten layer that reshapes the input in a 1D tensor. Similarly to [11], we combine the latent 1D features with a multi-layer perceptron (MLP) head up to dimensionality N_F. Conversely, to [7] that concatenates a single fully connected layer head, we propose a 3-layer MLP head (2 hidden layers with ReLU). The output of this contracting path is used to extract the queries q and the dictionary keys k, which are used to train both encoders. Similarly to [5], we propose integrating instance normalization in the backbone and the contracting path instead of batch normalization. Instance normalization performs better when the training set lacks high variability, as it can be the radar sounder case.

3. RESULTS AND CONCLUSIONS

We successfully validated the proposed method on data acquired in Greenland by MCoRDS-3 in 2017 that image the ice sheet up to the bedrock (more details on the dataset and the preprocessing are in [1, 6]). As the network's backbone, we considered the DeepLabV3+ network that extracts $N_F = 128$ features. We trained the network for 400 epochs for $k = 1$ and 200 epochs for $k > 1$. We fixed a batch size of 8 patches and reduced the learning rate starting from an initial value of $l_r = 0.02$. We set $t = 2$ as we segment radargrams in $N_{\omega=3}$ classes as in [3, 6]– ω_{IS} is englacial layers, ω_N is the noise-limited regions (thermal noise, echo-free zone, and free space above the surface), and ω_B is the basal interface/bedrock. The qualitative results in Fig. 4 show a radargram patch segmented at $t = 1$ in signal and noise-limited regions and at $t = 2$ in ice targets, bedrock, and noise-limited regions. Although the results show a good segmentation performance, the basal area is poorly represented.

The quantitative results are calculated in terms of overall accuracy, sensitivity (the probability that a pixel is correctly classified), and specificity (the probability that a pixel is incorrectly classified) for unambiguous pixels. At $t = 1$, the overall accuracy is about 86%, the sensitivity is 0.7787 and 0.9822, and the specificity is 0.9822 and 0.0.7787 for the signal and noise-limited regions classes, respectively. At



(a) (b) $t = 1$ (c) $t = 2$

Fig. 4. Example of a radargram patch (a) that is segmented in – signal in white and noise-limited regions in black (b) at $t = 1$, and in icy targets in white, bedrock in yellow, and noise-limited regions in black (c) at $t = 2$.

$t = 2$, the overall accuracy is about 88,74%, the sensitivity is equal to 0.8660, 0.8378, and 0.9585, and the specificity is 0.9628, 0.8691, and 0.7485 for the icy class, the bedrock, and noise-limited regions classes, respectively. Hence, the proposed method can discriminate more classes with increasing incremental learning steps.

4. CONCLUSIONS AND FUTURE WORKS

The paper proposes a method based on an incremental learning strategy that segments new classes with increasing incremental learning steps. The training strategy exploits a proxy task and contrastive learning to enhance the quality of the semantic features extracted by the network. We also propose to distill the knowledge of previous incremental learning steps to avoid catastrophic forgetting. The results validated the proposed method in segmenting MCoRDS-3 radargrams acquired on Greenland, considering new detailed classes at each incremental step. The incremental steps are critically linked to the number of classes in the defined hierarchy and the level of detail (number of targets) desired for the segmentation map.

In future works, we plan to use a more complex class hierarchy of the subsurface targets and analyze the error propagation in the segmentation maps.

5. ACKNOWLEDGMENTS

The Italian Space Agency supported this work through the" Missione JUICE - Attività dei team scientific dei Payload per Lancio, commissioning, operazioni e analisi dati" under Contract Agenzia Spaziale Italian- Istituto Nazionale di Astrofisica (ASI-INAF) and Contract 2023-6-HH.0. We acknowledge the use of data from CReSIS generated with support from the University of Kansas, NASA Operation IceBridge grant NNX16AH54G, NSF grants ACI-1443054, OPP-1739003, and IIS-1838230, Lilly Endowment Incorporated, and Indiana METACyt Initiative.

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